CHPTER # 7

Asmptotes:

=> Asi line jisko koi curve touch kr rahi ho

TYPES:

Three types of asymptote

=> Horizontal asymptote
=> verticle asymptote
=> oblique or inclined asymptote

Rules OF FINDING ASMPTOTES

=> First of all arrange the given equation of curve into descending order

Horizontal Asymptotes:-

Equate coefficients of highest power of x to zero if any

Verticle Asymptotes:-

Equate coefficients of highest power of y to zero.

Oblique or Inclined asymptotes:-

Equation of inclined asymptote is

$$y = mx + c$$

Put $x=1$, $y=m$ in the highest degree term and equate to zero

i.e $\phi_n(m) = 0$

$\Rightarrow m = m_1, m = m_2, m = m_3 \cdots m = m_n$

Value of c

$$c = \frac{\phi_{n-1}(m)}{\phi_n'(m)}$$

For a single value of m

Value of c in the Presence of two equal values of m:-

By putting value of m in the below formula we will get the value of c

$$\frac{c^2}{2!}\phi_n''(m) + \frac{c}{1!}\phi_{n-1}'(m) + \phi_{n-2}(m) = 0$$

c for three equal values of m

$$\frac{c^3}{3!}\phi_n'''(m) + \frac{c^2}{2!}\phi_{n-1}''(m) + \frac{c}{1!}\phi_{n-2}'(m) + \phi_{n-3}(m) = 0$$

Ex # 7.1

Find equations of the asymptotes:

1 $y = \dfrac{(x-2)^2}{x^2}$

$$x^2 y = (x-2)^2$$
$$x^2 y - (x-2)^2 = 0$$
$$x^2 y - (x^2 + 4 - 4x) = 0$$
$$x^2 y - x^2 + 4x - 4 = 0$$

Horizontal Asymptotes

coefficient of highest power of $x = 0$

$$y - 1 = 0$$

$$\boxed{y = 1}$$

which is horizontal asymptotes

Verticle Asymptote:

coefficient of highest power of $y = 0$

$$x^2 = 0 \qquad \boxed{x = 0}$$

which is verticle

The required assymptotes are

$y = 1$ $\quad x = 0$

2 $x^2y^2 = 12(x-3)$

$x^2y^2 - 12x + 48 = 0$

Horizontal assymptote:

coefficient of highest power of : $x = 0$

$y^2 = 0$

$\boxed{y = 0}$ which horizontal assymptote

Verticle assymptote:

coefficient of heighest Power of $y = 0$

$x^2 = 0$

$\boxed{x = 0}$

Hence there will be no inclined assymptote

3. $2xy = x^2 + 3$

$x^2 - 2xy + 3 = 0$

Horizontal assymptote

As the highest degree term of x has coefficient 1. so there will be No Horizontal Asymtote

Vertical assymptote

coefficient of highest degree term of y is $= -2x$

$-2x = 0$

$\boxed{x = 0}$ which is V.A

Inclined Assymptote

After arranging

$$x^2 - 2xy - 3 = 0$$

$$\phi_2(x,y) = x^2 - 2xy = 0$$

Put $x = 1$, $y = m$

$$1 - 2m = 0$$

$$\boxed{m = \frac{1}{2}}$$

Value of c for $m = 1/2$

Formula $c = -\dfrac{\phi_{n-1}(m)}{\phi_n'(m)}$

$$c\,\phi_n'(m) + \phi_{n-1}(m) = 0$$

Now $c\,\phi_2(m) + \phi_1(m) = 0$ — 1

Here $\phi_2(m) = 1 - 2m$

$\phi_2'(m) = -2$

$\phi_1(m) = 0$

using values in (1)

$c(-2) + 0 = 0$

$c = 0$

equation $y = mx + c$

$$\boxed{y = \frac{1}{2}x}$$ Inclined Assymptotes

Required assymptotes are

$x = 0$ $y = \frac{1}{2}x$ are required

assymptotes of curve

$x^2(x-y)^2 + a^2(x^2-y^2) = a^2xy$

$x^2(x^2+y^2-2xy) + a^2x^2 - a^2y^2 = a^2xy$

$x^4+x^2y^2-2x^3y+a^2x^2-a^2y^2-a^2xy=0$ — (1)

Horizontal assymptote:

As coefficient of highest power $x = 1$ then there will be Horizontal assymptotes

vertical assymptote

coefficient of highest power of $y = x^2+a^2$

$x^2-a^2 = 0$

$\boxed{x = \pm a}$ which is V.A

Inclined or oblique:

Arranging in decending order

$x^4+x^2y^2-2x^3y+a^2x^2-a^2y^2=a^2xy=0$

Here $\phi_4(x,y) = x^4+x^2y^2-2x^3y$

Put $x=1$ $y=m$

$\phi_4(m) = 1+m^2-2m=0$

$1+m^2-2m=0$

$m^2-2m+1=0$

$(m-1)^2=0$

$(m-1)(m-1)=0$

$m=1$ $m=1$

value of c for $m=1,1$

Formula = $\frac{c^2}{2!}\phi_n''(m) + \frac{c^1}{1!}\phi_{n-1}'(m) + \phi_{n-2}(m)$

$\frac{c^2}{2!}\phi_4''(m) + \frac{c}{1!}(\phi_3'(m)) + \phi_2(m) = 0$ ...2

Here $\phi_4(m) = 1 + m^2 - 2m$

$\phi_4'(m) = 2m - 2$

$\phi_4''(m) = 2$

$\phi_3(m) = 0$

$\phi_2(x,y) = a^2x^2 - a^2y^2 + a^2xy$

Put $x = 1 \quad y = m$

$\phi_2(m) = a^2 - a^2m^2 - a^2m$

$\phi_2(m) = a^2(1 - m^2 - m)$

Now Putt values in formula

$\frac{c^2}{2}\,2 + c(0) + a^2(1 - m^2 - m) = 0$

$c^2 + a^2(1 - m^2 - m) = 0$

$c^2 + a^2(1 - m^2 - m) = 0$

$c^2 + a^2(1 - 1 - 1) = 0$

$c^2 - a^2 = 0$

$(c + a)(c - a) = 0$

$(c + a) = 0 \qquad (c - a) = 0$

$c = -a \qquad c = a$

Now for $c = -a, a$ inclined assym is $y = mx + c$

$y = mx + c$

$\boxed{y = x - a}$

$y = mx + c$

$\boxed{y = x + a}$

Hence required asymptotes are

$x=a$ $\quad x=-a$ $\quad y=x-a$ $\quad y=x+a$

5. $(x-y)^2(x^2+y^2) - 10(x-y)x^2 + 12y^2 + 2x + y = 0$

$(x^2-y^2-2xy)(x^2+y^2) - 10x^3 + 10x^2y + 12y^2 + 2x + y = 0$

$x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3 - 10x^3 + 10x^2y + 12y^2 + 2x + y = 0$ ——— 1

Horizontal assymptote

As coefficients of H.P $x=1$

So there will be no Horizontal A

Verticle assymptote

coefficient of H.P $y=1$

So there will be no vertical A

Inclined assymptote:

$x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3 - 10x^3 + 10x^2y + 12y^2 + 2x + y = 0$

Here $\phi_4(x,y) = x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3$

Put $x=1$, $y=m$

$\phi_4(m) = 1 + m^2 - 2m + m^2 + m^4 - 2m^3$

$m^4 - 2m^3 + 2m^2 - 2m + 1 = 0$

$(m^2)^2 - 2m^3 + 2m^2 - 2m + 1 = 0$

$(m^2+1)^2 - 2m(m^2+1) = 0$

$(m^2+1)(m^2+1-2m) = 0$

$(m^2+1)(m-1)^2 = 0$

$(m^2+1) = 0$ $\quad (m-1)^2 = 0$

$m = \pm i$ Neglected $\quad (m-1)^2 = 0$

Value of c for $m = 1, 1$

Formula:

$$\frac{c^2}{2!}\phi_n''(m) + \frac{c^1}{1!}\phi_{n-1}'(m) + \phi_{n-2}(m) = 0$$

$$\frac{c^2}{2!}\phi_4''(m) + c^1\,\phi_3'(m) + \phi_2(m) = 0 \quad \text{— (2)}$$

$$\phi_4(m) = m^4 - 2m^3 + 2m^2 - 2m + 1$$

$$\phi_4'(m) = 4m^3 - 6m^2 + 4m - 2$$

$$\phi_4''(m) = 12m^2 - 12m + 4$$

$$\phi_3(x, y) = 10x^3 + 10x^2y$$

Put $x = 1$ $y = m$

$$\phi_3(m) = 10(m - 1)$$

$$\phi_3'(m) = 10$$

$$\phi_2(m) = 12m^2$$

Now Put values in (2)

$$\frac{c^2}{2}(12m^2 - 12m + 4) + c(10) + 12m^2 = 0$$

$$c^2(6m^2 - 6m + 2) + 10c + 12m^2 = 0$$

Now Put $m = 1$

$$c^2(6 - 6 + 2) + 10c + 12 = 0$$

$$2c^2 + 10c + 12 = 0$$

$$c^2 + 5c + 6 = 0$$

$$c^2 + 3c + 2c + 6 = 0$$

$$c(c + 3) + 2(c + 3) = 0$$

$$(c + 2)(c + 3) = 0$$

$$c + 2 = 0$$

$c = -2$ $c = -3$

equation of inclined assymptote

$y = mx + c$ $\because c = -2, -3$

$y = x - 2$ $y = x - 3$

Hence required asymptotes are

$y = x - 2$, $y = x - 3$

6. $x^2y + xy^2 + xy + y^2 + 3x = 0$

Horizontal assymptote

Putting coefficient of highest Power term x equal to zero

$y = 0$ which is H.A

Verticle assymptote

coefficient of highest Power term $y = 0$

$x + 1 = 0$

$x = -1$ which is V.A

Inclined assymptote:

$x^2y + xy^2 + xy + y^2 + 3x = 0$

Now $\phi_3(x, y) = x^2y + xy^2$

Put $x = 1$ & $y = m$ equate to zero

$\phi_3(m) = m + m^2$

$\phi_3(m) = m(1+m) = 0$

$m(1+m) = 0$

$m = 0$ $\quad 1+m = 0$

$m = -1$

Value of c for $m = -1$

Formula $c = -\frac{\phi_{n-1}(m)}{\phi_n'(m)}$

$$c = -\frac{\phi_2(m)}{\phi_3'(m)} \quad \text{---------} \quad (*)$$

Here $\phi_3(m) = m + m^2$

$\phi_3'(m) = 1 + 2m$

$\phi_2(x,y) = xy + y^2$

$x = 1 \qquad y = m$

$\phi_2(m) = m + m^2$

Now put values in (*)

$$c = -\frac{m + m^2}{1 + 2m}$$

Put value of $m = -1$

$$\boxed{c = 0}$$

For $m = 0$ c will also $c = 0$

Equation of asymptote

$y = mx + c$

$y = -mx$

Hence Required asymptotes are

$y=0 \quad x=-1 \quad , \quad y=-x$

7. $(x-y+1)(x-y-2)(x+y) = 8x-1$

$(x^2-xy-2x-xy+y^2+2y+x-y-2)(x+y)=8x-1$

$(x^2-y^2-2xy-x+y-2)(x+y) = 8x-1$

$x^3+xy^2-2x^2y-x^2+xy-2x+x^2y+y^3-2xy^2-xy+y^2-2y = 8x-1$

$x^3+y^3-xy^2-x^2y-x^2+y^2-10x-2y+1=0$

H.A

As coefficient of highest Power of $x=1$, Hence there is no H.A

V. A

As coefficient of highest Power of $y=1$, Hence there is no V.A

Inclined . A

$\phi_n(x,y) = 0$

$\phi_3(x,y) = x^3+y^3-xy^2-x^2y = 0$

Put $x=1$, $y=m$

$\phi_3(m) = 1+m^3-m^2-m = 0$

$m^3+m^2-m+1 = 0$

$m^2(m-1)-1(m-1) = 0$

$(m^2-1)(m-1) = 0$

$(m-1)(m+1) = 0$

$(m-1) = 0, \quad (m+1) = 0$

$m = 1 \qquad m = -1 \qquad m = 1$

value of c for $m = 1$

Formula: $\frac{c^2}{2!}\phi_n''(m) + \frac{c}{1!}\phi_{n-1}'(m) + \phi_{n-2}(m) = 0$

$\frac{c^2}{2!}\phi_3''(m) + \frac{c}{1!}\phi_2'(m) + \phi(m) = 0$ — (*)

$\phi_3(m) = m^3 - m^2 - m + 1$

$\phi_3'(m) = 3m^2 - 2m - 1$

$\Rightarrow \phi_3''(m) = 6m - 2$

$\phi_2(x, y) = x^2 + y^2$

Put $x = 1, \quad y = m$

$\Rightarrow \phi_2(m) = -1 + m^2$

$\phi_2'(m) = 2m$

$\phi_1(m) = -10x - 2y$

Put $x = 1 \quad y = m$

$\Rightarrow \phi_1(m) = -10 - 2m$

Put values in (*)

$\frac{c^2}{2!}(6m-2) + c(2m) - 10 - 2m = 0$

Put value of $m = 1$

$\frac{c^2}{2}(4) + 2c - 2 - 10 = 0$

$2c^2 + 2c - 12 = 0 \Rightarrow c^2 + c - 6 = 0$

$(c+3)(c-2) = 0$

$c + 3 = 0 \qquad c - 2 = 0$

$c = -3$ $\quad$ $c = 2$

equation of assymptote

$y = mx + c$

$y = x - 3$ $\quad$ $y = x + 2$

Value of c for $m = -1$

Formula $\quad c = -\dfrac{\phi_2(m)}{\phi_3'(m)}$

$$c = -\frac{(-1 + m^2)}{3m^2 - 2m - 1} = \frac{1 - m^2}{3m^2 - 2m - 1}$$

$c = 0$

equation of assymptote

$y = mx + c$ $\quad$ $y = -x$

Hence Required assymptotes are

$y = x - 3$ $\quad$ $y = x + 2$ $\quad$ $y = -x$

8. $y^3 + x^2y + 2xy^2 - y + 1 = 0 \longrightarrow (1)$

H. A

Put coefficient of H.P of $x = 0$

$y = 0$

V.A

As coefficient of Highest power of $y = 0$ So $1 \neq 0$

Hence there is no V.A

$y^3 + x^2y + 2xy^2 - y + 1 = 0$

Put $x = 1$ and $y = m$ in highest power tem of 1 and equate to zero

$\phi_3(x,y) = y^3 + x^2y + 2xy^2 = 0$

$\phi_3(m) = m^3 + m + 2m^2 = 0$

$\phi_3(m) = m^3 + 2m^2 + m = 0$

$m(m^2 + 2m + 1) = 0$

$m = 0$ $\quad (m+1)^2 = 0$

$m + 1 = 0 \quad m + 1 = 0$

$m = -1 \quad m = -1$

Value of c fr $m = -1$

Formula: $\frac{c^2}{2!}\phi_3''(m) + c\phi_2'(m) + \phi_1(m)$

$\phi_3(m) = m^3 + 2m^2 + m$

$\phi_3'(m) = 3m^2 + 4m$

$\phi_3''(m) = 6m + 4$

$\phi_2(m) = 0 \qquad d_1(m) = -y$

$\phi_1(m) = -m$

$\frac{c^2}{2!}(6m+4) + c^2(0) + (-m)$

$c^2(3m+2) - m$

Put value of $m = -1$

$c^2(3(-1)+2) - (-1)$

$-c^2 + 1 = 0$

$c = \pm 1$

$c = 1$, $c = -1$

Equation of assymptote

$y = mx + c$

where $m = -1$ $c = 1$

$y = -x + 1$

For $c = -1$

$y = -x - 1$

Hence Required assymptotes are

$y = 0$ $y = -x + 1$ $y = -x - 1$

9. $y(x-y)^2 = x+y$

$y(x^2 + y^2 - 2xy) = x + y$

$x^2y + y^3 - 2xy^2 - x - y = 0$ — (1)

Horizontal assymptote:

Put coefficient of H.P of $x = y$

Put $y = 0$

which H.A

Vertical assymptote:

Put coefficient of H.P of $y = 0$

$1 \neq 0$

Hence there is no V.A

Inclined .A

$x^2y + y^3 - 2xy^2 - x - y = 0$

Put highest Power of 1 $= y = mx = 1$

$\phi_3(x,y) = x^2y + y^3 - 2xy^2 = 0$

$\phi_3(m) = m + m^3 - 2m^2 = 0$

$m=0$ $\quad (m-1)^2 = 0$

$m=0$ $\quad (m-1) = 0$

$m = 1$ $\quad m = 1$

Value of c for $m = 1$

Formula: $\frac{c^2}{2!}\phi_3''(m) + \frac{c^2}{1!}\phi_2'(m) + \phi_1(m) = 0$ A.II

$\phi_3(m) = m^3 + m - 2m^2$

$\phi'(m) = 3m^2 + 1 - 4m$

$\phi''(m) = 6m - 4$

$\phi_2(m) = 0$

$\phi_1(x,y) = -1 - y$ A.V

Put $x=1$, $y=m$

$\phi_1(x,y) = -1 - m$

Put these values in formula

$\frac{c^2}{2}(6m-4) - 1 - m = 0$ A.I

$c^2(3m-4) - 1 - m = 0$

Put value of m

$c^2(3(1)-4) - 1 - 1 = 0$

$c^2 - 2 = 0$

$c = \pm\sqrt{2}$

$c = \sqrt{2}$ $\quad c = -\sqrt{2}$

Equation of assymptote

$y = mx + c$

$y = x + \sqrt{2}$

$y = x - \sqrt{2}$

10. $x^2y^2 - x^2y - 3x^2$ $\quad (a+b)^3 = a^3+b^3+3a^2b+3ab^2$

10. $x^2y^2(x^2-y^2)^2 = (x^2+y^2)^3$

$x^2y^2(x^4+y^4-2x^2y^2) - (x^2+y^2)^3 = 0$

$x^6y^2 + x^2y^6 - 2x^4y^4 - (x^6+y^6+3x^4y^2+3xy^4) = 0$

$x^6y^2 + x^2y^6 - 2x^4y^4 - x^6 - y^6 - 3x^4y^2 - 3xy^4 = 0$

H.A

coefficient of H.P of $x = y^2 - 1$

Put $y^2 - 1 = 0$

$y = \pm 1$ which is H.A

V.A

coefficient of H.P of $y = x^2 - 1$

$x^2 - 1 = 0 \quad x = \pm 1$

$x = \pm 1$ which is V.A

I.A

Put $x = 1$ and $y = m$ in H.P degree term of 4 and equate it to zero

$\phi_6(x,y) = x^6y^2 + x^2y^6 - 2x^4y^4$

$\phi_6(m) = m^2 + m^6 - 2m^4 = 0$

$m^2 - 2m^4 + m^6 = 0$

$m^2(m^4 - 2m^2 + 1) = 0$

$m^2 = 0$ $\quad (m^2-1)^2 = 0$

$m = 0$ $\quad m^2 - 1 = 0$ $\quad m^2 - 1 = 0$

$m = 0$ $\quad m = \pm 1$ $\quad m = \pm 1$

Value of c for m = 1

Formula: $\frac{c^2}{2!}\phi_8''(m) + c\,\phi_7'(m) + \phi_6(m) = 0$

$\phi_8(x,y) = x^6y^2 + x^2y^6 - 2x^4y^4$

$\phi_8(m) = m^6 - 2m^4 + m^2$

$\phi_8'(m) = 6m^5 - 8m^3 + 2m$

$\phi_8''(m) = 30m^4 - 24m^2 + 2$

$\phi_7(m) = 0$

$\phi_6(x,y) = -x^6 - y^6 - 3x^4y^2 - 3x^2y^4$

Put $x = 1$, $y = m$ and equate to zero

$\phi_6(m) = -1 - m^6 - 3m^2 - 3m^4 = 0$

Put values in formula (*)

$\frac{c^2}{2!}(30m^4 - 24m^2 + 2) + c(0) + (-1 - m^6 - 3m^2 - 3m^4) = 0$

Put $m = 1$

$c^2(15(1) - 12(1)^2 + 2) + (-1 - (1)^6 - 3(1) - 3(1))$

$4c^2 - 8 = 0$

$c^2 = 2$

$c = \pm\sqrt{2}$

equation of assymptote

$y = mx + c$

$m = 1$ $\quad c = \pm\sqrt{2}$

$y = x + \sqrt{2}$ $\quad y = x - \sqrt{2}$

value of c for $m = -1$

Put $m = -1$ in (*)

$c^2(15(-1)^4 - 12(-1) + (-1) + (-1 - (-1)^6 - 3(-1)^2 - 3(1)^4)$

$c = \pm\sqrt{2}$

Equation of assymptote

$y = mx + c$

$m = -1$ $\quad c = \pm\sqrt{2}$

$y = -x + \sqrt{2}$ $\quad y = -x - \sqrt{2}$

Hence required assymptotes are

$y = \pm 1$, $x = \pm 1$, $y = -x + \sqrt{2}$ $\quad y = -x - \sqrt{2}$ $\quad y = x - \sqrt{2}$

11 $xy^2 = (x+y)^2$

$xy^2 = x^2 + y^2 + 2xy$

$xy^2 - x^2 - y^2 - 2xy = 0$

H.A

coefficient of H.P of $x = -1$

$-1 \neq 0$ Hence there is no H.A

V.A

A coefficient of H.P of $y = x - 1$

Put $x - 1 = 0$

$\boxed{x = 1}$ which is Horizontal assymptote

I.A

The highest degree term

$\phi_3(x, y) = xy^2$

Put $x = 1$ $\quad y = m$

$\phi_3(m) = m^2$

$1 - m^2 = 0$ not $m = 0$

So there is no assymptote

12. $xy^2 - x^2y - 3x^2 - 2xy + y^2 + x - 2y + 1 = 0$

H.A

coefficient of H.P of $x = -y - 3$

Put $-y - 3 = 0$

$\boxed{y = -3}$ which is H.A

V.A

coefficient of H.P of $y = x + 1$

Put $x + 1 = 0$

$\boxed{x = -1}$ which is V.A

I.A

Highest degree term

$\phi_3(x, y) = xy^2 - x^2y$

$\phi_3(m) = m^2 - m$

$m^2 - m = 0$

$m(m - 1) = 0$

$m = 0 \qquad m = 1$

value of c for $m = 1$

Fomula $= -\dfrac{\phi_{n-1}(m)}{\phi_n'(m)}$

$\phi_3(m) = -\dfrac{\phi_2(m)}{\phi_3'(m)}$

$\phi_2(x, y) = -3x^2 - 2xy + y^2$

Put $x = 1 \quad y = m$

$= -3 - 2m + m^2$

$\phi_3'(m) = 2m - 1$

Put values in formula

$$c = \frac{-(-3 - 2m + m^2)}{2m - 1}$$

$$c = \frac{-(-3 - 2(1) + (1)^2}{2(1) - 1}$$

$$c = -(-4) = 4$$

equation of assymptote

$y = mx + c$

$m = 1 \qquad c = 4$

$$\boxed{y = x + 4}$$

Hence Required assymptotes are

$y = -3 \qquad x = -1 \qquad y = x + 4$

WORKING OF POLAR CURVE:-

let $r = f(\theta)$ be the curve

put $r = \infty$ in the equation of the curve and find value of θ

Say $\theta = \alpha, \beta, \gamma, \ldots$

Then by using formula

$$p = r\sin(\alpha - \theta)$$

we can find the equation of assymptote

where

$$P = \lim_{\theta \to a, \beta, \gamma} r^2 \frac{d\theta}{dr}$$

13. $r = \frac{a}{\theta}$ — 1 where $r = \infty$

$\frac{a}{\theta} = 0$ $\quad a = 0$ $\quad$ So $\alpha = 0$

diff 1 w.r.t θ

$$\frac{dr}{d\theta} = a\left[\frac{\theta(0) - (1)(1)}{\theta^2}\right]$$

$$\frac{dr}{d\theta} = -\frac{a}{\theta^2}$$

$$\frac{d\theta}{dr} = -\frac{\theta^2}{a}$$

Multiply b.sides by r^2

$$r^2\frac{d\theta}{dr} = -r^2\frac{\theta^2}{a}$$

Take limit

$$\lim_{\theta \to 0} r^2\frac{d\theta}{dr} = \lim_{\theta \to 0}\left[-\frac{a^2}{\theta^2} \cdot \frac{\theta^2}{a}\right]$$

$$P = \lim_{\theta \to 0} (-a)$$

$$P = -a$$

Formula $P = r\sin(\alpha - \theta)$

$$-a = r\sin(0 - \theta)$$

$$-a = -r\sin\theta$$

$a = r\sin\theta$ Required

14. $r = \frac{a}{\sqrt{\theta}}$ ——— (1)

Put $r = \infty$ in 1

$\frac{a}{\sqrt{\theta}} = \infty$ $\sqrt{\theta} = 0$

$\theta = 0$ i.e $a = 0$

Now diff 1 w.r.t θ

$$\frac{dr}{d\theta} = a\left[\frac{\theta.(0) - (1)\,1/2\sqrt{\theta}}{\theta}\right]$$

$$\frac{dr}{d\theta} = a\left[\frac{-1/2\sqrt{\theta}}{\theta}\right]$$

$$\frac{dr}{d\theta} = \frac{-1}{2}\frac{a}{\theta^{3/2}}$$

$$\frac{d\theta}{dr} = -\frac{2\theta^{3/2}}{a}$$

multiply b.sides by r^2

$$\frac{r^2 d\theta}{dr} = -\frac{r^2 2\theta^{3/2}}{a}$$

Take limits

$$\lim_{\theta \to 0} \frac{r^2 d\theta}{dr} = \lim_{\theta \to 0}\left[-r^2 \frac{2\theta^{3/2}}{a}\right]$$

$$\lim_{\theta \to 0} \frac{r^2 d\theta}{dr} = \lim_{\theta \to 0}\left[-\frac{a^2}{r\sqrt{\theta}} \cdot \frac{2\theta^{3/2}}{a}\right]$$

$$\lim_{\theta \to 0} \frac{r^2 d\theta}{dr} = \lim_{\theta \to 0} -2a\sqrt{\theta}$$

$P = 0$

Formula of assymptote

$P = r\sin(\alpha - \theta)$

$0 = r\sin(0 - \theta)$

$\sin\theta = 0$

$\theta = \sin^{-1}(0)$

$\theta = 0$ which is Required.

15. $r = a\sec\theta + b$ ——— (1)

$r = \frac{a}{\sin\theta} + b$

$r = \frac{a + b\sin\theta}{\sin\theta}$ ⟶ (2)

⟶ Step

diff (1) w.r.t θ

$\frac{dr}{d\theta} = a(-\operatorname{cosec}\theta\cot\theta)$

$\frac{d\theta}{dr} = \frac{1}{a\operatorname{cosec}\theta\cot\theta}$

multiply b.sides by r^2

$r^2\frac{d\theta}{dr} = -r^2$

$r^2\frac{d\theta}{dr} = -\left[\frac{a + b\sin\theta}{\sin\theta}\right]^2 \frac{1}{a\operatorname{cosec}\theta\cot\theta}$

$= \frac{a^2 + b\sin^2\theta}{\sin^2\theta} \cdot \frac{\sin\theta\sin\theta}{a\cos\theta}$

$r^2\frac{d\theta}{dr} = \frac{(a + b\sin\theta)^2}{a\cos\theta}$

Put $r = \infty$

$\sin\theta = 0$

$\theta = \sin^{-1}(0)$

$\theta = 0, \pi$

i.e $\alpha = 0$

$\beta = \pi$

value of P when $a=0$

Take limit

$$\lim_{\theta\to 0} \frac{r^2 d\theta}{dr} = \lim_{\theta\to 0} -\left[\frac{(a+b\sin\theta)^2}{a\cos\theta}\right]$$

$P = -a$

Formula of assymptote ~~$y = mx + c$~~ $P = r\sin(\alpha-\theta)$

$-a = r\sin\theta(0-\theta)$

$-a = -r\sin\theta$

$a = r\sin\theta$ Required assymptote.

(2)

value of P when $\beta = \pi$

$$\lim_{\theta\to\pi} \frac{r^2 d\theta}{dr} = \lim_{\theta\to\pi} -\left[\frac{(a+b\sin\theta)^2}{a\cos\theta}\right]$$

$$P = -\frac{(a^2)}{a} = a$$

Formula $P = r\sin(\beta-\theta)$

$a = r\sin(\pi-\theta)$

$a = r\sin\theta$ $\quad \sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

which is Required assymptote

16. $r = 2a\sin\theta\tan\theta$

$r = 2a\sin\theta \cdot \frac{\sin\theta}{\cos\theta}$

$r = \frac{2a\sin^2\theta}{\cos\theta}$ —— (1)

Put $r = \infty$

$\cos\theta = 0$ $\quad \theta = \cos^{-1}(0)$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ $\quad$ i.e $\quad \alpha = \frac{\pi}{2}$ $\quad \beta = \frac{3\pi}{2}$

Now diff 1 w.r.t θ

$$\frac{dr}{d\theta} = 2a\left[\frac{\cos\theta(2\sin\theta\cos\theta) + \sin^2\theta(\sin\theta)}{(\cos\theta)^2}\right]$$

$$\frac{dr}{d\theta} = 2a\left[\frac{2\sin\theta\cos^2\theta + \sin^3\theta}{\cos^2\theta}\right]$$

$$\frac{d\theta}{dr} = \frac{\cos^2\theta}{2a(2\sin\theta\cos^2\theta + \sin^3\theta)}$$

multiply b. sides r^2

$$r^2\frac{d\theta}{dr} = r^2\,\frac{\cos^2\theta}{2a(2\sin\theta\cos^2\theta + \sin^3\theta)}$$

$$r^2\frac{d\theta}{dr} = \frac{4a^2\sin^4\theta}{\cos^2\theta}\cdot\frac{\cos^2\theta}{2a(2\sin\theta\cos^2\theta + \sin^3\theta)}$$

$$r^2\frac{d\theta}{dr} = \frac{2a\sin^4\theta}{\sin\theta(2\cos^2\theta + \sin^2\theta)}$$

$$r^2\frac{d\theta}{dr} = \frac{2a\sin^3\theta}{2\cos^2\theta + \sin^2\theta} \quad \text{---} \quad (*)$$

value of P for $\alpha = \pi/2$

$$\lim_{\theta\to\pi/2} r^2\frac{d\theta}{dr} = \lim_{\theta\to\pi/2} \frac{2a\sin^3\theta}{2\cos^2\theta + \sin^2\theta}$$

$P = 2a(1)$

$P = 2a$

by using formula. $P = r\sin(\alpha - \theta)$

$2a = r\sin(\pi/2 - \theta)$

$2a = r\cos\theta$

value of P For $\beta = \frac{3\pi}{2}$ in *

$$\lim_{\theta \to 3\pi/2} r^2 \frac{d\theta}{dr} = \lim_{\theta \to 3\pi/2} \frac{2a\sin^3\theta}{2\cos^2\theta + \sin^2\theta}$$

$P = -2a$

By using Formula $P = r\sin(\beta - \theta)$

$-2a = r\sin\left(\frac{3\pi}{2} - \theta\right)$

$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$-2a = -r\cos\theta$

$2a = r\cos\theta$ Required assymptote

17. $r = r\sin 2\theta = a\cos 3\theta$

$r = \frac{a\cos 3\theta}{\sin 2\theta}$ ——— 1

Put $r = \infty$

$\sin 2\theta = 0$

$2\theta = \sin^{-1}(0)$

$2\theta = 0, \pi$

$\theta = 0, \pi/2$ i.e $\alpha = 0$ $\beta = \frac{\pi}{2}$

Now diff w.r.t θ

$$\frac{dr}{d\theta} = a\left[\frac{\sin 2\theta(-3\sin 3\theta) - \sin 3\theta\, 2\cos 2\theta}{(\sin 2\theta)^2}\right]$$

$$\frac{d\theta}{dr} = \frac{\sin^2 2\theta}{a(-3\sin 2\theta \sin 3\theta - 2\sin 3\theta \cos 2\theta)}$$

multiply r^2 on b.sides

$$r^2\frac{d\theta}{dr} = r^2 \frac{\sin^2 2\theta}{a(-3\sin 2\theta \sin 3\theta - 2\sin 3\theta \cos 2\theta)}$$

$$r^2\frac{d\theta}{dr} = \frac{a^2\cos^2 3\theta}{\sin^2 2\theta} \cdot \frac{\sin^2 2\theta}{a(-3\sin 2\theta \sin 3\theta - 2\sin 3\theta \cos 2\theta)}$$

$$r^2\frac{d\theta}{dr} = -\frac{a\cos^2 3\theta}{3\sin 2\theta \sin 3\theta + 2\sin 3\theta \cos 2\theta}$$

Take limit at b.sides

$$\lim_{\theta \to 0} r^2\frac{d\theta}{dr} = \lim_{\theta \to 0} -\frac{a\cos^3 3\theta}{3\sin 2\theta \sin 3\theta + 2\sin 3\theta \cos 2\theta}$$

$$P = \frac{-a}{0 + 2(1)(1)} = \frac{-a}{2}$$

equation of assymptote

$$P = r\sin(\alpha - \theta)$$

$$\frac{-a}{2} = r\sin(0 - \theta)$$

$$\frac{-a}{2} = -r\sin\theta$$

$$a = 2r\sin\theta \quad \text{Required.}$$

value of P for $\beta = \pi/2$

$$\lim_{\theta \to \pi/2} \frac{r^2 d\theta}{dr} = \lim_{\theta \to \pi/2} \frac{-a\cos^2 3\theta}{3\sin 2\theta \sin 3\theta + 2\sin 3\theta \cos 2\theta}$$

$P = 0$

by Formula

$P = r\sin(\beta - \theta)$

$0 = r\sin\left(\frac{\pi}{2} - \theta\right)$

$r\cos\theta = 0$

$r\cos\theta = 0$

$\cos\theta = 0$

$\theta = \cos^{-1}(0)$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ i.e $\alpha = \pi/2$, $\beta = \frac{3\pi}{2}$

Required Formula

$a = 2r\sin\theta$ $\theta = \frac{\pi}{2}$ $\theta = \frac{3\pi}{2}$

18. $r = \frac{a}{1 - \cos\theta}$ ____ 1

Put $r = \infty$

$\frac{a}{1-\cos\theta} = \infty$

$1 - \cos\theta = 0$

$\cos\theta = 1$

$\theta = \cos^{-1}(1)$

$\theta = 0$ i.e $\alpha = 0$

Now diff 1 w.r.t θ

$$\frac{dr}{d\theta} = a.\frac{(1-\cos\theta).0 - a(\sin\theta)}{(1-\cos\theta)^2}$$

$$\frac{dr}{d\theta} = \frac{-a\sin\theta}{(1-\cos\theta)^2} \Rightarrow \frac{d\theta}{dr} = -\frac{(1-\cos\theta)^2}{a\sin\theta}$$

multiply b.sides by r^2

$$r^2\frac{d\theta}{dr} = -r^2\frac{(1-\cos\theta)^2}{a\sin\theta}$$

$$r^2\frac{d\theta}{d\theta} = -\frac{a^2}{(1-\cos\theta)^2}\cdot\frac{(1-\cos\theta)^2}{a\sin\theta}$$

$$\lim_{\theta\to 0} r^2\frac{d\theta}{dr} = \lim_{\theta\to 0} -\frac{a}{\sin\theta}$$

$$P = -\infty$$

There is no assymptote for curve

A. $r\sin n\theta = a$

$$r = \frac{a}{\sin n\theta} \quad \text{---------- 1}$$

Now Put $r = \infty$

$$\frac{a}{\sin n\theta} = \infty$$

$$\sin n\theta = 0$$

$$n\theta = \sin^{-1}(0)$$

$$n\theta = k\pi \qquad k = 0, 1, 2, 3$$

$$\theta = \frac{k}{n}\pi \quad \text{i.e} \quad \alpha = \frac{k}{n}\pi$$

Now diff $\frac{1}{r}$ w.r.t θ

$$\frac{dr}{d\theta} = a\left[\frac{(\sin n\theta).0 - n\cos n\theta}{(\sin n\theta)^2}\right]$$

$$\frac{dr}{d\theta} = -\frac{a\,n\cos n\theta}{(\sin n\theta)^2}$$

$$\frac{d\theta}{dr} = -\frac{(\sin n\theta)^2}{an\cos n\theta}$$

multiply r^2 on b.sides

$$\frac{r^2 d\theta}{dr} = -r^2\frac{(\sin n\theta)^2}{an\cos n\theta}$$

$$\frac{r^2 d\theta}{dr} = -\frac{a^2}{(\sin n\theta)^2}\cdot\frac{(\sin n\theta)^2}{an\cos n\theta}$$

$$\lim_{\theta \to \frac{k\pi}{n}} \frac{r^2 d\theta}{dr} = -\lim_{\theta \to \frac{k\pi}{n}} \frac{a}{n\cos n\theta}$$

$$P = \frac{-a}{n\cos n\left(\frac{k\pi}{n}\right)}$$

$$P = \frac{-a}{n\cos k\pi}$$

$$P = \frac{-a}{n}\sec k\pi$$

By using Formula

$$P = r\sin(a-\theta)$$

$$-\frac{a}{n}\sec k\pi = -r\sin\left[\frac{k\pi}{n} - \theta\right]$$

$$\frac{a}{n}\sec\frac{k\pi}{n} = r\sin\left(\theta - \frac{k\pi}{n}\right)$$

Required assymptote

20. $r(e^\theta - 1) = a(e^\theta + 1)$

$$r = \frac{a(e^\theta+1)}{e^\theta-1} \quad \text{— 1}$$

Put $r = \infty$

$e^\theta - 1 = 0$

$e^\theta = 1$

$e^\theta = e^0$ $\quad \theta = 0$ i.e $a = 0$

Now diff 1 w.r.t θ

$$\frac{dr}{d\theta} = a\left[\frac{(e^\theta-1)e^\theta - (e^\theta+1)e^\theta}{(e^\theta-1)^2}\right]$$

$$\frac{dr}{d\theta} = a\left[\frac{e^{2\theta} - e^\theta - e^{2\theta} - e^\theta}{(e^\theta-1)^2}\right]$$

$$\frac{dr}{d\theta} = \frac{-2ae^\theta}{(e^\theta-1)^2} \Rightarrow \frac{d\theta}{dr} = \frac{(e^\theta-1)^2}{2ae^\theta}$$

multiply r^2 b. sides

$$\frac{r^2 d\theta}{dr} = -r^2\frac{2ae^\theta}{(e^\theta-1)^2}$$

$$r^2\frac{d\theta}{dr} = -\frac{a^2(e^\theta+1)^2}{(e^\theta-1)^2}\cdot\frac{(e^\theta-1)^2}{2ae^\theta}$$

$$r^2\frac{d\theta}{dr} = -\frac{a(e^\theta+1)^2}{2e^\theta}$$

$$\lim_{\theta\to 0} r^2\frac{d\theta}{dr} = \lim_{\theta\to 0} \frac{-a(e^\theta+1)^2}{2e^\theta}$$

$$P = \frac{-a(4)}{2} = \frac{-4a}{2}$$

$$P = -2a$$

by Formula $P = \pm r\sin(a-\theta)$

$$-2a = r\sin(0-\theta)$$

$$-2a = -r\sin\theta$$

$2a = r\sin\theta$ Required

21. $r^n \sin n\theta = a^n$

$$r^n = \frac{a^n}{\sin n\theta} \quad \text{—} \quad (1)$$

Put $r = \infty$

$$\sin n\theta = \frac{a^n}{\infty} = 0$$

$$\sin n\theta = 0$$

$$n\theta = \sin^{-1}(0) \Rightarrow k\pi$$

$$\theta = \frac{k\pi}{n} \quad \text{i.e} \quad a = \frac{k\pi}{n}$$

Now diff 1 w.r.t θ

$$nr^{n-1}\frac{dr}{d\theta} = a^n\left[\frac{\sin n\theta(0) - n\cos n\theta}{\sin^2 n\theta}\right]$$

$$nr^{n-1}\frac{dr}{d\theta} = \frac{-a^n n\cos n\theta}{\sin^2 n\theta}$$

$$\frac{dr}{d\theta} = \frac{-a^n \cos n\theta}{r^{n-1} \sin^2 n\theta}$$

$$\frac{d\theta}{dr} = \frac{-r^{n-1} \sin^2 n\theta}{a^n \cos n\theta}$$

multiply b.side by r^2

$$r^2 \frac{d\theta}{dr} = -r^2 \cdot \frac{r^{n-1} \sin^2 n\theta}{a^n \cos n\theta}$$

$$r^2 \frac{d\theta}{dr} = -r^{n+1} \frac{\sin^2 n\theta}{a^n \cos n\theta}$$

$$= -r^n \cdot r \frac{\sin^2 n\theta}{a^n \cos n\theta}$$

$$r^2 \frac{d\theta}{dr} = -\frac{a^n}{\sin n\theta} \times \frac{a}{\sin^{1/n} n\theta} \times \frac{\sin^2 n\theta}{a^n \cos n\theta}$$

$$r^2 \frac{d\theta}{dr} - \frac{a (\sin n\theta)^{2-1-\frac{1}{n}}}{\cos n\theta} = \frac{-a \sin n\theta^{1-\frac{1}{n}}}{\cos n\theta}$$

$$\lim_{\theta \to \frac{k\pi}{n}} r^2 \frac{d\theta}{dr} = \lim_{\theta \to k\pi/n} \frac{-a \sin n\theta^{1-\frac{1}{n}}}{\cos n\theta}$$

$$P = \frac{-a \left(\sin n\left(\frac{k\pi}{n}\right)\right)^{1-\frac{1}{n}}}{\cos n\left(\frac{k\pi}{n}\right)}$$

$$= \frac{-a(\sin k\bar{\pi})^{1-\frac{1}{n}}}{\cos k\bar{\pi}} \quad \because k = 1, 2, 3, 4 \ldots$$

$$p = 0$$

by using Formula

$$p = r\sin(\alpha - \theta)$$

$$0 = r\sin\left(\frac{k\bar{\pi}}{n} - \theta\right)$$

$$\sin\left(\frac{k\bar{\pi}}{n} - \theta\right) = 0$$

$$\frac{k\bar{\pi}}{n} - \theta = \sin^{-1}(0)$$

$$\frac{k\bar{\pi}}{n} - \theta = 0$$

$$\theta = \frac{k\bar{\pi}}{n}$$ Required asymptote

22. $r^2\sin\theta = a^2\cos 2\theta$

$$r^2 = \frac{a^2\cos 2\theta}{\sin\theta} \quad \text{---} \quad 1$$

Put $r = \infty$ in 1

$$\sin\theta = 0$$

$$\theta = \sin^{-1}(0)$$

$\theta = 0, \pi$ $\quad \alpha = 0 \quad \beta = \bar{\pi}$

Now diff 1 w.r.t θ

$$2r\frac{dr}{d\theta} = a^2\left[\frac{-\sin\theta\, 2\sin 2\theta - \cos 2\theta\cos\theta}{\sin^2\theta}\right]$$

$$= a^2\left[\frac{-2\sin\theta\sin 2\theta - \cos\theta\cos 2\theta}{\sin^2\theta}\right]$$

$$\frac{dr}{d\theta} = -\frac{a^2}{\sin^2\theta}\cdot\frac{1}{2r}(2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta)$$

$$\frac{d\theta}{dr} = -\frac{2r\sin^2\theta}{a^2(2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta)}$$

$$r^2\frac{d\theta}{dr} = -r^2\cdot\frac{2r\sin^2\theta}{a^2(2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta)}$$

$$= -r^3 \times \frac{2\sin^2\theta}{a^2(2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta)}$$

$$r^2\frac{d\theta}{dr} = -\frac{a^3(\cos 2\theta)^{3/2}}{\sin^{3/2}\theta}\cdot\frac{2\sin^2\theta}{a^2(2\sin\theta\sin 2\theta + \cos\theta\cos$$

$$r^2\frac{d\theta}{dr} = \frac{-2a\sin^{1/2}\theta\cos^{3/2}2\theta}{2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta}$$

Now value of P when $a=0$

$$\lim_{\theta\to 0} r^2\frac{d\theta}{dr} = \lim_{\theta\to 0}\frac{2a\sin^{1/2}\theta\cos^{3/2}2\theta}{2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta}$$

$P = 0$

by using Formula

$$P = r\sin(\alpha - \theta)$$

$$0 = r\sin(0 - \theta)$$

$$0 = -r\sin\theta$$

$$\sin\theta = 0$$

$\theta = 0$ which is Required basic angle

Now value of P when $\beta = \pi$

$$P = \lim_{\theta \to \pi} \frac{-2a\sin^{1/2}\theta\cos^{3/2}2\theta}{2\sin\theta\sin 2\theta + \cos\theta\cos 2\theta}$$

$P = 0$

$\boxed{\theta = 0}$ after using formula

Required $\theta = 0$